Correction and Clarification

Neural Moving Horizon Estimation for Robust Flight Control
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Dear Readers,

In Eqs. (15a) and (34), the gradient cross term is printed as

$$\sum_{k=t-N}^{t} \mathrm{Tr}\left(\boldsymbol{W}_k^T \boldsymbol{L}_k^{wx} \boldsymbol{X}_k\right)$$

The correct upper summation limit is t − 1, so the term should read

$$\sum_{k=t-N}^{t-1} \mathrm{Tr}\left(\boldsymbol{W}_k^T \boldsymbol{L}_k^{wx} \boldsymbol{X}_k\right)$$

The process-noise sensitivities are defined only for stages $k = t - N, \cdots, t - 1$.
We also clarify the statement of linear computational complexity in the MHE horizon N. For p independent learnable parameters and fixed state and noise dimensions, the complete gradient trajectory using our algorithm requires $O(Np)$ arithmetic. This reduces to $O(N)$ when p is horizon-independent (e.g., with stage-varying weights parameterized by a fixed-dimensional $\boldsymbol{\Theta}$ and differentiated directly with respect to $\boldsymbol{\Theta}$, as implemented using the sparse parameterization introduced in Section IV-C and Algorithm 2) but becomes $O(N^2)$ when all stage weights are independent and $p = O(N)$. These clarifications do not affect the formulation of the auxiliary MHE system, the Kalman-filter-based recursive solver, or the reported results.

We apologize for any confusion.

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